

FPGA-Based Echo State Network for Battery SOC/SOH Estimation

Consolidated Results Summary

System Overview

Architecture: Time-multiplexed reservoir computing (ESN), single physical neuron core sequentially reused across all reservoir neurons.

Parameter	Value
Reservoir size (N_RES)	100 neurons
Input dimensionality (N_IN)	4 features
Arithmetic format	Q6.10 signed fixed-point
Target device	Xilinx Artix-7, xc7a100tcsg324-1
Target board	Digilent ArtyA7-100T
Input features	voltage_measured, current_measured, temperature_measured, current_load

Reservoir initialization (weights freshly generated, not recovered from an original training run documented explicitly, seed=42):

- $W_{in} \sim U(-0.5, 0.5)$, dense
- W : 10% connection density, 80% excitatory / 20% inhibitory (magnitude $U(0,1)$), spectral radius rescaled to 0.9
- Bias: zero

Hardware Verification

Check	Result
Bit-exact Python golden model vs. Vivado/XSim (MAC, bias, sum, tanh_in, tanh_out)	Exact match, confirmed independently multiple times across the full pipeline
X-corruption (unknown/uninitialized values) across real-data simulation runs	Zero occurrences, across $32 \text{ cycles} \times 6,400$ rows each
Timestep-dependent input addressing ($u_addr = \text{timestep} \times N_IN + \text{win_count}$)	Verified correct across all 64-timestep windows

Dataset

Source: NASA Prognostics Center of Excellence Li-ion battery dataset, batteries B0005, B0006, B0007, B0018.

Split: Matches the generalization framing of the reference literature, train on one cell, test on unseen cells:

- **Train:** B0005
- **Test:** B0006, B0007, B0018 (never used for feature-scaling or hyperparameter selection)

Coverage:

Battery	Role	Cycles used	SOH range covered
B0005	Train	0, 28, 56, 84, 105, 126, 146, 167	1.000 $\rightarrow$ 0.710
B0006	Test	0, 28, 56, 83, 105, 126, 146, 167	1.000 $\rightarrow$ 0.583
B0007	Test	0, 28, 56, 83, 105, 126, 146, 167	1.000 $\rightarrow$ 0.758
B0018	Test	0, 22, 44, 65, 82, 98, 115, 131	1.000 $\rightarrow$ 0.723

Total: 32 cycles $\times$ 64 timesteps = 2,048 labeled reservoir-state rows.

Labeling methodology:

- **SOC:** within-cycle Coulomb counting against each cycle's measured discharge capacity
- **SOH:** capacity fade relative to each battery's own first discharge cycle
- **Features:** Min-Max normalized, scaler fit on B0005 (train) only, applied to all batteries, avoids train/test leakage

SOC / SOH Estimation Results (FPGA-derived reservoir state + ridge-regression readout)

Readout: ridge regression, regularization selected via leave-one-cycle-out cross-validation within the training battery only (test batteries never used for tuning).

Target	Split	MAE	RMSE	R ²
SOC	Train (B0005)	0.0118	0.0151	0.9663
SOC	Test (B0006/7/18, unseen)	0.0220	0.0289	0.8783
SOH	Train (B0005)	0.0420	0.0554	0.7367
SOH	Test (B0006/7/18, unseen)	0.0614	0.0851	0.4766

Interpretation:

- SOC estimation generalizes well to entirely unseen battery cells.
- SOH estimation shows genuine, above-baseline predictive signal, but is limited by training diversity (only 8 distinct SOH values seen during training, from a single battery). This is disclosed as a limitation, not overstated.

Baseline Comparison - Extended Kalman Filter (SOC only)

A 1RC Thevenin equivalent-circuit EKF was implemented as a classical baseline, fit exclusively on B0005 (OCV polynomial + R_0 via least squares; R_1/C_1 fixed to physically reasonable values after a data-driven fit degenerated, disclosed limitation, discharge-only data provides no rest-voltage segments to fully identify RC dynamics). Evaluated on the identical 32 cycles used for the ESN comparison, same train/test split, no label leakage (SOC initialized to 1.0 at cycle start, a physical fact of the test protocol, not a leaked label).

Method	Test MAE	Test RMSE	Test R^2
EKF (1RC Thevenin)	0.0311	0.0387	0.7818
FPGA ESN readout	0.0220	0.0289	0.8783

Key Finding: The FPGA ESN outperforms the EKF baseline on all three metrics, on identical data and evaluation methodology.

Caveat: The EKF here is a simplified baseline (no true rest-OCV characterization available in this dataset); a more heavily-tuned EKF could close part of this gap.

FPGA Synthesis Results

Resource utilization (post-implementation, xc7a100tcsg324-1):

Resource	Used	Available	Utilization
Slice LUTs	77	63,400	0.12%
Slice Registers	66	126,800	0.05%
Block RAM Tiles	7.5	135	5.56%
DSP48 Slices	0	240	0.00%

Notable: The entire arithmetic datapath (multiply-accumulate, tanh activation) synthesizes into LUT fabric with zero DSP48 usage. BRAM usage is dominated by the 100×100 reservoir weight matrix (w_bram, 5 of 7.5 tiles).

Power (implemented netlist, vectorless/activity-derived estimate):

Metric	@ 100 MHz	@ 250 MHz
Total on-chip power	0.099 W	0.101 W
Dynamic	0.001 W (2%)	0.004 W (4%)
Static (device leakage)	0.097 W (98%)	0.097 W (96%)

Timing:

Clock target	Result
100 MHz (10.000 ns)	Closes, WNS = 6.264 ns
250 MHz (4.000 ns)	Closes, WNS = 0.783 ns
Estimated true Fmax	$\approx$ 310 MHz

Zero failing endpoints on setup, hold, or pulse-width checks at both tested frequencies.

Latency / Throughput

Directly measured from simulation (100 MHz reference clock): **42,807 cycles** per full 100-neuron reservoir update (428 cycles/neuron).

Clock frequency	Time per reservoir update (SOC/SOH)
100 MHz (demonstrated)	428.1 μ s
250 MHz (confirmed)	171.2 μ s
$\sim$ 310 MHz (estimated Fmax)	138.1 μ s

Real-time context: Typical BMS voltage/current sampling occurs at 1–100 Hz. Even at the conservative 100 MHz clock, one full reservoir inference (428.1 μ s) leaves:

Sample rate	Headroom @ 100 MHz
1 Hz	2,336×
10 Hz	234×
100 Hz	23×

The design is substantially over-provisioned for real-time BMS deployment at its most conservative demonstrated clock speed.

Summary of Contributions

- Fixed-point (Q6.10) ESN reservoir architecture, verified bit-exact between an independent Python golden model and Vivado/XSim.
- Time-multiplexed single-neuron-core design achieving <0.2% LUT/FF utilization and zero DSP48 usage on a 100-neuron reservoir.
- Evaluation on real NASA battery data (4 cells), proper train/unseen-test split, disclosed methodology throughout.
- Direct, same-data comparison against a classical EKF baseline, with the FPGA ESN outperforming on SOC estimation.
- Full synthesis characterization: resource utilization, power, timing closure up to 250 MHz and real-time throughput margin analysis.

Known Limitations (for explicit disclosure in the paper)

- **SOH estimation accuracy:** ($R^2=0.48$ test) is limited by training-set diversity (8 SOH values from one battery); more training cycles across more batteries would likely improve this.
- **EKF baseline identification:** The EKF baseline's $R1/C1$ parameters could not be cleanly identified from discharge-only data (no rest periods); reported EKF numbers should be read as a simplified baseline, not a best-possible EKF implementation.